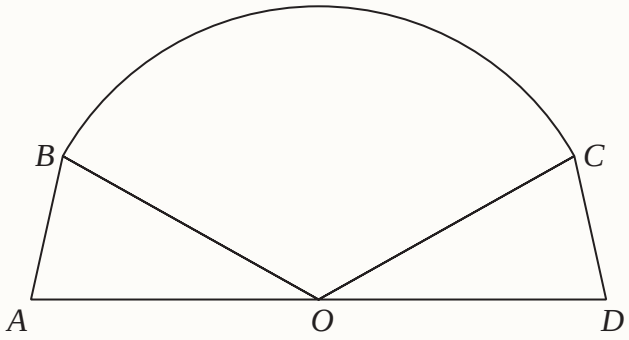


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the plan view for the design of a stage.

The design consists of a sector *OBC* of a circle, with centre *O*, joined to two congruent triangles *OAB* and *ODC*.

Given that

- angle *BOC* = 2.4 radians
- area of sector *BOC* = 40 m^2
- *AOD* is a straight line of length 12.5 m

(a) find the radius of the sector, giving your answer, in m, to 2 decimal places,

(2)

(b) find the size of angle *AOB*, in radians, to 2 decimal places.

(1)

Hence find

(c) the total area of the stage, giving your answer, in m^2, to one decimal place,

(3)

(d) the total perimeter of the stage, giving your answer, in m, to one decimal place.

(4)